

Exploring the Gender Divide in Real Estate Ownership and Property Tax Compliance

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Abstract

This paper investigates gender disparities in residential property ownership and tax compliance in a large Argentine municipality using detailed tax administrative data. While ownership is evenly distributed between women, men, and co-owned properties up to the 40th percentile of the value distribution, higher-value properties exhibit significant gender disparities, with women's share dropping to less than 20% in the top 1%. Tax compliance increases with property value, with an average evasion rate of 46%, and men and women are equally likely to meet their tax obligations across the distribution. However, women face slightly higher effective tax rates due to owning lower-value

properties, which are disproportionately affected by a mildly regressive tax schedule. Gender responses to enforcement measures are also comparable. A soft randomized communication campaign significantly increased timely payments equally for both men and women, with men responding more quickly. Similarly, the findings show no gender-based differences in responses to macroeconomic shocks such as COVID-19. The study underscores the role of property tax in promoting equitable revenue mobilization and highlights the importance of gender-disaggregated data for informing tax policy and enforcement strategies.

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1 Introduction

The property tax is increasingly recognized as a key component of equitable domestic revenue mobilization, with significant untapped potential for generating revenue (World Bank, 2021; Benitez, Mansour, Pecho, and Vellutini, 2023; Dom, Custers, Davenport, and Prichard, 2022; Maloney, Zambrano, Vuletin, Beylis, and Garriga, 2024). In many countries, real estate emerges as a primary repository of wealth, positioning property taxation as a potential tool for redistribution. However, as this tax grows in importance, taking into consideration its gender dimension is a central concern (Scot, Flores Ibarra, Moura, Feinmann, and Rocha, 2023; Komatsu, Ambel, Koolwal, and Yonis, 2021). Despite its relevance, the relationship between property taxation and gender outcomes remains largely understudied.

This paper provides new gender-focused empirical evidence on property ownership and taxation using administrative tax micro-data from *Tres de Febrero*, a large urban municipality in Argentina with over 360,000 inhabitants and about 120,000 households. Residents from Tres de Febrero are required to pay a monthly tax on their real estate, locally known as *Tasa por Servicios Generales* (TSG), which accounts for most of the local revenues in Argentine municipalities. We leverage rich tax administrative data and exogenous variation to show how gender interacts with several fundamental issues: the distribution of property ownership, tax evasion, and tax enforcement.

The analysis unfolds in three stages. First, we examine the distribution of residential ownership, presenting novel descriptive evidence on property ownership across the municipality, with a particular focus on gender disparities and the assessed values of properties. A significant challenge for such analyses is that property ownership records often lack explicit information on owners' gender. However, a unique feature of our setting is the structure embedded within national personal tax identifiers, which can be used to infer gender. Following methodological guidance from our team, the municipality's staff utilized this structural feature to infer the gender assigned at birth and the owners' ages for over 100,000 properties annually.

Our analysis reveals significant gender disparities in residential property ownership, primarily at the higher end of the property value distribution, where women own a smaller share than men. At the lower end of the property value distribution, up to the 40th percentile, ownership is fairly balanced, with women, men, and co-owned properties each accounting for about one-third of the residences. However, this balance shifts markedly as property values rise. While men's ownership share remains relatively stable throughout the distribution, women's share declines sharply for higher-value properties. This disparity is most pronounced in the top 1% of property values, where co-owned properties account for 50%, men own about 30%, and women's

ownership drops below 20%. These findings highlight the stark gender disparities in high-value property ownership, which are important to consider in the wealth distribution and gender equity literature.

Second, we conduct a detailed analysis of property tax compliance by gender and find that men and women are equally likely to pay their property tax bills. A key advantage of studying property taxes is that evasion is directly observed: owners are billed based on cadastral valuation, and we observe whether they pay or not. We characterize tax payment rates for women and men, examining how these rates vary across the wealth distribution, proxied by the properties' assessed value. We document substantial evasion, with slightly lower rates for women-owned properties (46%) compared to men-owned (48.5%).¹ The property tax gap is economically significant, amounting to roughly 8% of total municipal spending or the equivalent of the entire local public safety budget. On-time payment rates increase with property value, ranging from about 35% in the first percentile to 60% in the top percentile, though they remain similar for men and women across the distribution. However, women face slightly higher effective tax rates than men, driven by ownership of lower-value properties and a *de jure* regressive tax schedule. We also document a small but statistically significant difference in payment methods: women are slightly less likely to use electronic means to pay property taxes, even after controlling for age.

Third, we examine gender-specific responses to (soft) tax enforcement measures. We estimate the effects of personalized tax letters on current, past, and subsequent property tax payments for women and men by leveraging a large-scale field experiment conducted by Cruces, Tortarolo, and Vazquez-Bare (2023) in October 2020.² The intervention, launched amid the COVID-19 pandemic, sought to address a sharp decline in property tax payments driven by economic disruptions and lockdown measures. It consisted of sending a one-page personalized letter to randomly selected residences, providing information on the October 2020 bill (amount due and due date), past due debt, and online or in-person payment instructions.

Our causal analysis indicates that men and women respond similarly to personalized property tax letters. First, the letter recipients are substantially more likely to pay the October 2020 tax bill on time compared to those in the control group, with no statistically significant gender differences. For women, payment rates increased by 4.2 percentage points, a 12% rise from the control baseline of 35%. For men, the increase was 4.7 percentage points, representing a 13.5% improvement over their control baseline. The granularity of our data also allows us to examine

1. The average delinquency rates align with those reported for other Latin American economies (e.g., see Brockmeyer, Estefan, Ramírez Arras, and Suárez Serrato, 2021; Kresch, Walker, Best, Gerard, and Naritomi, 2023; Kapon, Del Carpio, and Chassang, 2024; Castro and Scartascini, 2015).

2. Cruces et al. (2023) use the campaign to study spillovers in property tax compliance across treated and untreated neighborhoods, but do not explore gender-specific effects.

payment behavior on a daily basis. Notably, men reacted more quickly to the tax letters, with the treatment effect appearing earlier and stabilizing after the due date. In contrast, treated women continued to make overdue payments in the days following, rapidly closing the initial gender gap in compliance. Second, the impact of the personalized tax letters extended beyond the targeted October 2020 bill, but the response is similar for men and women. Despite receiving the letters in early October, the intervention prompted some treated taxpayers to clear past-due debt from previous months and also pay their November and December bills. Third, while women present homogeneous responses across quintiles of property valuation, men exhibit a slightly negative gradient. Notably, in the lowest quintile, the personalized tax letter boosted timely payments of men by 8 percentage points—twice the increase observed for women.

Lastly, the inclusion of the recent pandemic in our analysis period allows us to examine gender-specific responses to a macroeconomic shock.³ Although payment rates dropped significantly—by approximately 25 percentage points—at the onset of the COVID-19 pandemic, we observe no noticeable gender-based differences in response. This indicates that macroeconomic shocks seem to affect property tax delinquency behavior similarly for both men and women.

Overall, our findings suggest that gender-targeted strategies may have limited potential for improving tax compliance in the context of property taxes, as men and women exhibit similar baseline payment rates and respond in comparable ways to personalized tax letters.⁴ Notwithstanding, the pronounced gender disparities in higher-value property ownership documented in our analysis raise important questions about wealth inequality and underscore the need for more research into the underlying causes and broader economic implications (e.g., differences in credit access).

We contribute to several strands of the literature studying asset ownership and the link between wealth taxation (including property taxes) and gender. Many countries recently highlighted the need to improve the collection of gender-disaggregated data on taxation in general, and on men’s and women’s property and capital ownership in particular, which is less commonly available (OECD, 2022). We add to a growing literature documenting gender patterns in property ownership and wealth (Holden and Tilahun, 2020; Kotikula and Raza, 2021; Komatsu et al., 2021; Gaddis, Lahoti, and Swaminathan, 2022; Scot et al., 2023), corroborating previous findings that women own fewer properties than men, particularly as property values increase. We document a substantial wealth gap, with women’s ownership shares declining significantly for higher-valued properties.

3. For recent work on gender and business cycles, see Gomes (2024).

4. However, we acknowledge that our study does not evaluate the effectiveness of gender-specific interventions implemented elsewhere, such as Awasthi, Pyle, Aggarwal, and Rakhimova (2023).

We also provide new evidence on gender differences in tax compliance. Prior research indicates that women are generally more compliant with tax filing and reporting obligations compared to men (Kleven, Knudsen, Kreiner, Pedersen, and Saez, 2011; Asmare and Yimam, 2020; Cabral, Gemmell, and Alinaghi, 2021). Several key factors have been proposed to explain this disparity, including women’s higher risk aversion (Croson and Gneezy, 2009), stronger tax morale, disparities in tax declaration and enforcement processes (Ambel and Woldeyes, 2024), and gender-specific social norms that interact with gaps in literacy, employment type, and access to information, among others (Komatsu, Shaukat, and Ozer, 2024). Our granular administrative data enables us to offer gender-disaggregated compliance insights on property taxation, in a low-enforcement setting, finding no meaningful gender differentials. Our finding contrasts with earlier studies, which suggested that women experienced either lower effective tax rates due to exemptions for lower-valued properties—more commonly owned by women in São Paulo, Brazil (Scot et al., 2023)—or, as Komatsu et al. (2021) found, a heavier tax incidence for female-headed households based on survey data from rural Ethiopia.

Lastly, while there is ample evidence on the effect of nudges—such as deterrence letters—on tax compliance and collection (Antinyan and Asatryan, 2024), research linking compliance and gender remains limited (see, for example, López-Luzuriaga and Scartascini, 2023; Awasthi et al., 2023). Hence, the results from this project exhibit great promise of informing policy debates and the conduct of tax policy that contribute toward the achievement of gender equality goals.

The remainder of the paper is organized as follows. In section 2, we describe the institutional context. Section 3 details the sources and composition of our tax data. Section 4 characterizes residential property ownership in *Tres de Febrero* by gender. Section 5 brings three stylized facts on property tax compliance and also leverages experimental evidence on differences in compliance responses across genders. Finally, Section 6 discusses some policy implications behind our results and then concludes.

2 Institutional Context

Tres de Febrero is an urban municipality (county) in the Greater Buenos Aires metro area, Argentina. As of 2022, it had approximately 365,000 residents and 115,000 households, making up 1.2% of the total population of Argentina (INDEC, 2022). It has the third highest population density in the province of Buenos Aires, with great connectivity and accessibility to neighboring areas, and contributes 2.4% to the total urban tax valuation of the province, ranking 13th among the 135 municipalities in fiscal valuation.

The municipality levies a local tax known as TSG (*Tasa por Servicios Generales*), which is linked to the provision of lighting and cleaning services. This type of tax is present in every municipality of the Buenos Aires Province and is the main source of locally collected resources (Porto, Fernández Felices, and Puig, 2019). Tres de Febrero is no exception, as the TSG accounts for 20% of its total resources and 45% of its own revenue in 2021.

The monthly TSG tax comprises both variable and fixed components. In our period of analysis 2018–2020, the tax liability for residential taxpayer i was calculated as follows:

$$\text{Monthly tax}_i = \left[\text{Cadastral value}_i \times \text{Tax rate}/12 + \text{Fixed charge} \right] \quad (1)$$

The variable component is calculated by applying a tax rate to the property’s assessed value. These tax rates vary across eight property-use categories defined by the municipality: residential, industrial, commercial, wholesale establishments, mixed-use residential with commerce or factory, empty lots, civil entities, and religious entities. As of 2021, the statutory tax rates ranged from 0.32% to 2.48% (see Table 1). Property assessments are based on the cadastre maintained by the Revenue Agency of the Province of Buenos Aires (ARBA), though the municipality reserves the right to update these values based on its own criteria.

The fixed charge corresponds to a flat fee for specific municipal services, including security, health, fire departments, and maintenance of public spaces. Notably, fixed charges constitute a larger share of the total tax burden for lower-value properties. This share varies significantly across the property value distribution, comprising 65% of the total tax burden for properties in the first decile but only 14% for those in the tenth decile (Figure C.4). As a result, the overall tax schedule has historically exhibited a mildly regressive structure, as illustrated in Figure 6.⁵

TSG bills are issued monthly and are due in the first weeks of each month, with twelve equal installments per year. Alternatively, residents can opt for an advance annual payment at the beginning of the year instead of monthly installments. While cadastral values are seldom updated, both the tax rates and fixed charges are adjusted annually to keep up with inflation.

Property Tax Enforcement. Enforcing local property taxes in developing countries has historically been a tall order. Anecdotal evidence from conversations with public servants at Tres de Febrero’s tax agency reveals several factors limiting their ability to enforce the TSG property tax:

5. While this approach simplifies revenue collection and ensures a baseline contribution toward municipal services (e.g., see Bergeron, Tourek, and Weigel, 2024), it reflects archaic tax regimes that may have been suitable in earlier periods but conflict with modern views of tax fairness, which often advocate for progressive tax systems that impose a larger relative burden on wealthier households (Chancel, Piketty, Saez, and Zucman, 2022). The municipality is currently reforming this structure by recalibrating the balance between fixed and variable components to enhance equity.

1. **Limited human, technical, and legal resources.** Subnational tax agencies often struggle with enforcement due to inadequate IT systems and limited human resources. In 2020, for instance, Tres de Febrero’s tax audit department operated with just five employees overseeing 130,000 TSG payers and over 4,000 businesses subject to a local turnover tax (TISH). This team relied on data from a 1980s IBM mainframe to determine taxpayer debts and payments, leading to high costs for identifying debts and noncompliance, making large-scale audits infeasible. Given their limited resources, local tax agencies typically focus their enforcement efforts on a few large businesses that generate most of the income and tax revenue, rather than overseeing tens of thousands of households. For example, in 2020 ten companies accounted for half of the business tax revenue, compared to 25,000 properties that made up half of TSG’s revenue.
2. **High political cost.** The political economy of enforcing a local property tax is contentious (Cabral and Hoxby, 2012), especially in high-inflation contexts. In the Province of Buenos Aires, municipal elections occur biennially, and mayors tend to avoid comprehensive tax enforcement campaigns on delinquent households during odd years to maintain voter support. Moreover, mayors have been reluctant to update the cadastral value of properties (i.e., the tax base) for the same reason (Christensen and Garfias, 2021). Consequently, the property tax’s share in the overall tax structure has declined over time.
3. **Financial disincentives to compliance.** Frequent debt regularization programs and a partial adjustment of delinquent tax debt can further disincentivize timely tax payments. Until 2022, the municipality charged an annual simple interest rate of 12% on late payments, significantly below the inflation rate. Additionally, the municipality has implemented payment plans with preferential interest rates and debt forgiveness (*moratorias*) in four of the last five years. These actions incentivize taxpayers to postpone payments to reduce the real value of their tax debt or even wait for the next *moratoria* to settle their obligations (e.g., see Lauletta and Montano Campos, 2023).

Collectively, these factors provide indicative evidence of low enforcement capacity for property taxes in developing countries. In this context, property tax compliance is essentially quasi-voluntary and potentially driven by intrinsic motives such as social norms and reciprocity (Luttmer and Singhal, 2014). Consequently, there is potential for soft interventions (e.g., tax information letters) to influence outcomes like tax delinquency. Furthermore, in this setting of imperfect enforcement, it is worth examining whether compliance rates differ between women and men.

3 Administrative Data

We use administrative data from January 2018 through December 2023, compiled through an ongoing collaboration with the local revenue agency in Tres de Febrero.

Local property tax data. The primary database consists of monthly property tax records constructed from the monthly bills issued to account holders. The unit of observation is an account that coincides with a dwelling unit. The data contain the following billing details: account number (unique property identifier), address, name of locality (neighborhood), year and month of the bill (12 yearly bills), the monthly fee (in pesos), a payment indicator, due date, date of payment, days overdue, means of payment (cash or electronic), type of account (residential, retailer, manufacturer), and also information about property size.

Property ownership, assessed value, and gender. Tres de Febrero receives a yearly data file from the Province of Buenos Aires containing the assessed values of local properties as well as individual tax identifiers of up to two owners. The authorities merge this register with their tax data using the cadastral nomenclature.⁶ Crucially, there is a way to infer the sex assigned at birth and the age of the owners based on the first two digits and the last digit of the individual tax identifier (as well as from owners' names). Appendix B provides a detailed description.

RCT data. We have access to treatment assignments from a large-scale randomized communication campaign conducted by Cruces et al. (2023), who estimate direct and spillover effects on property tax compliance. The campaign consisted of sending 25,000 personalized letters to randomly selected properties in October 2020 with reminders about due taxes, information about the status of the account, due dates, past due debt, and payment methods. We combine this database with the previous two sources to investigate heterogeneous behavioral responses to nudges between men and women and across the distribution of property values.

Summary Statistics. Table 2 provides an overview of property characteristics and taxation trends spanning the years 2018, 2019, and 2020. In Panel A, we show that residential properties constitute 83% of all properties in Tres de Febrero, with a concentration in the town of Caseros, where approximately 30% of all properties are situated. Panel B sheds light on the taxation dynamics and payment behavior within Tres de Febrero. In 2019, the mean assessed property value stood at 1.2 million pesos (about USD 30,000), with an average tax liability of 10,101 pesos (about USD 250). Between 2018 and 2020, the variable rate increased from 0.73% to 1.25%, while the empirical Effective Tax Rate (ETR), defined as total property taxes paid (including variable and fixed

6. The cadastral nomenclature is the number assigned to any property based on its location in the city, urbanization, street, and position in the street-block. This is an essential document issued by the Cadastre Office of the municipality of each locality.

components) as a share of assessed value, rose from 0.46% to 0.63%. In the same year, the ratio of tax compliance payments to tax liabilities at the intensive margin was 63%, with 68% of liabilities being paid on time. Notably, the proportion of consistent taxpayers (always payers) decreased by 15 percentage points from 55% to 40% in the period analyzed, while approximately one-third of total properties failed to comply with at least one payment during the year (never payers).

Figure 1 illustrates the spatial distribution of residential property valuation and tax compliance across *Tres de Febrero*. Darker blue shades indicate areas with higher average property values. Overlaid on these areas are pie charts representing tax compliance: green segments show the percentage of taxpayers who pay all twelve monthly installments ("always payers"), yellow segments indicate those who pay some bills ("sometimes payers"), and red segments represent those who do not pay any bills ("never payers"). The municipality exhibits remarkable heterogeneity in payment rates, which appears to correlate closely with properties' values. This heterogeneity prompts us to further explore the gender divide in real estate ownership and property tax compliance.

4 Real Estate Ownership and Gender

In the urban landscape of Tres de Febrero, our analysis reveals a notable gender disparity in property ownership, with women holding fewer properties than men across time and within higher property-assessed values. As detailed in Table 2, about 21% of the properties are solely owned by women, men own 51%, and 28% are co-owned by men-women, men-men, or women-women. Consequently, a substantial majority of properties fall under exclusive male ownership.

Figure 2 displays residential ownership by gender and percentiles of assessed value in 2018. We observe that property ownership shares are similar across women, men, and co-owned properties at the lower end of the property value distribution and differ markedly at the upper end. Up to the 40th percentile, each group owns one-third of the residential properties. While the share of men-owned properties remains constant across the distribution, the share of women-owned properties decreases substantially for higher-value properties. For the top 1% of property values, co-owned properties represent 50% of all ownership, men own around 30%, and women own less than 20% of residential properties. This trend is further illustrated in Figure 3, which shows the cumulative distribution of property values for men and women. Women's property ownership tends to be concentrated at the lower end of the value spectrum.

Our findings on gender-property ownership disparities are consistent with patterns observed in other contexts. For example, in São Paulo, Brazil, Scot et al. (2023) found that women own

about 30% of the properties up to the 80th percentile of the value distribution, while men own approximately 55%-60%, with the remainder being co-owned properties. Importantly, women's ownership rapidly declines at the upper end of the value distribution, reaching just 20% among the highest-valued properties, similar to the trend in Tres de Febrero. However, unlike Tres de Febrero, where co-owned properties are prevalent at the top percentile, in São Paulo, these are predominantly owned by men.

5 Property Tax Compliance and Gender

Are men and women equally tax-compliant? In this section, we first document property tax compliance facts by gender and then analyze their responses to a randomized tax information campaign. In a nutshell, we find that (1) men and women exhibit similar tax compliance levels across the value distribution, (2) women face slightly higher effective tax rates and contribute a smaller share of total tax revenue than men because they tend to own lower-valued houses, and (3) men and women respond similarly to enforcement campaigns.

5.1 Stylized Facts

Fact #1: There is a substantial tax gap, with slightly lower evasion for women-owned properties (46%) compared to men-owned (48.5%). Figure 4 compares aggregate property tax liabilities to the total tax paid by gender for residential properties in 2019. In that year, the total tax owed amounted to ARS 544 million, while the total tax actually paid was ARS 295 million, resulting in a tax gap of ARS 249 million (46% of the tax owed). This tax gap is economically significant, roughly equivalent to the local public safety expenditure of ARS 246 million, or about 8% of total spending (see Table A1). This gap is slightly higher for properties owned by men (48.5%) compared to those owned by women (46%) and co-owned properties (43%).

Fact #2: The on-time payment rate for property taxes increases with property value but is similar for women and men across the entire distribution. In 2019, 47% of residential property tax bills in *Tres de Febrero* were paid on time. We define an on-time payment as any bill settled by the 27th day of the due month, as well as bills paid in full for the entire year. Figure 5 illustrates how the share of on-time payments (out of the twelve in a year) varies across percentiles of assessed residential properties values, ranging from about 35% in the first percentile to 60% in the top percentile. Notably, payment rates are similar between women and men, highlighting a uniform approach to compliance regardless of gender. This finding adds to

the growing interest in understanding the distributional aspects of property tax compliance by gender.

Fact #3: Women face slightly higher effective tax rates than men, driven by ownership of lower-value properties and a *de jure* regressive tax schedule. Our dataset allows us to calculate the statutory ETR (tax owed as a fraction of property assessed value) and the empirical ETR (tax paid as a fraction of property assessed value). Three key insights emerge. First, as shown in [Figure 6](#), taxpayers at the lower end of the property value distribution face ETRs that are 2 to 3 times higher than those at the top, regardless of gender. This disparity arises from fixed charges of the tax schedule that disproportionately burden low-value properties. Second, due to high non-compliance, the empirical ETR is substantially lower than the statutory ETR across the distribution. While the schedule is *de jure* regressive, it becomes more proportional *de facto* when accounting for non-compliance. Third, as illustrated in [Figure 7](#), women face slightly higher average effective tax rates (both statutory and empirical) compared to men and co-owners. This difference is driven by women’s greater ownership of lower-value properties ([Figure 3](#)) and the mildly regressive nature of the tax schedule. Overall, these findings suggest that non-compliance and ownership patterns may undermine both vertical equity (tax disparities across the distribution) and horizontal equity (tax burden across gender).

Fact #4: Women are slightly less likely to use electronic methods to pay property taxes. [Table 3](#) shows differences in means from linear probability models using various payment methods as dependent variables. The first two columns document that, on average, women are about 2 percentage points more likely to pay all their property tax bills in cash (compared to a baseline of 60% who use cash). Conversely, they are less likely to use online payment methods. The result remains similar when we consider the fraction of tax payments made online, as shown in the last columns. These results hold whether we perform a simple comparison of means (odd columns) or include controls for assessed value, age, and geographical location (even columns), suggesting relatively small differences in payment method preferences between men and women.

5.2 A Randomized Tax Communication Campaign

Do men and women respond differently to tax enforcement measures? Tax administrations increasingly use communication tools such as letters, emails, and text messages to enhance tax compliance and streamline their collection strategies. We examine whether men and women exhibit different responses to such interventions. While the effect of taxpayer reminder letters on tax compliance and collection is well documented in existing, rigorous RCTs (Antinyan and Asatryan, 2024; Hjort, Moreira, Rao, and Santini, 2021; Mascagni, 2018), studies specifically ad-

addressing the interplay between compliance and gender remain scarce (López-Luzuriaga and Scartascini, 2023; Awasthi et al., 2023). Our analysis aims to fill this gap by analyzing gender-specific responses to tax enforcement measures, thereby contributing to a more nuanced understanding of effective tax administration strategies.

5.2.1 Experimental design

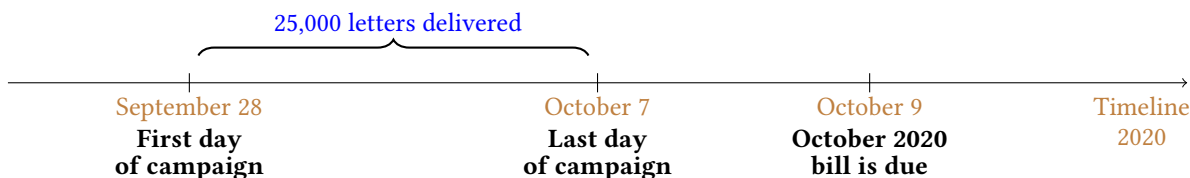
We estimate the effects of personalized tax letters on current and subsequent property tax payments for women and men by leveraging a large-scale field experiment conducted by Cruces et al. (2023) in *Tres de Febrero* in October 2020. The intervention consisted of sending a one-page personalized letter to randomly selected residences with information on the October 2020 bill (amount due and due date), past due debt, and how to pay online or in person.⁷ It was originally designed as a partial population experiment to detect the presence of social interactions (spillovers) in tax compliance.

Randomization took place in two stages—first at the street-block level, and then at the taxpayer (i.e., property) level.⁸ In the first stage, Cruces et al. (2023) randomly divided the sample of 3,982 street-blocks (clusters) into four groups with increasing treatment intensity (with 0%, 20%, 50%, and 80% of the accounts treated). In the second stage, they randomly selected about 25,000 accounts within the last three saturation groups to receive the treatment letter. To keep things simple, in this paper, we ignore the saturation groups and simply compare property tax payments between taxpayers who received a letter (treated) and those who did not (untreated).

The timeline of the intervention is displayed below. The letters were delivered between September 28th and October 7th, 2020, corresponding to payments due on October 9th, 2020, and past due debt (if any). The intervention was run on residential properties present in the municipality in 2019. The sample size consists of 68,808 accounts distributed in 3,982 blocks. The frequency of payments is highly polarized. About 45 percent of the accounts paid the twelve 2019 monthly bills (*always payers*), and about 35 percent did not pay any bill at all (*never payers*). The proportion of always payers is relatively low and, therefore, leaves room for potential behavioral responses from non-compliant and partially-compliant neighbors, and this was compounded by the context of the pandemic, during which lockdown measures reduced payments even from highly compliant individuals.

7. Figure C.7 in the appendix provides an anonymized example of the intervention letter.

8. For details on the experiment (randomization, power calculations, pre-analysis plan) see Cruces et al., 2023. The experimental design was preregistered in the AEA RCT Registry (RCT ID: **AEARCTR-0006569**).



5.2.2 Findings

We begin the empirical analysis by estimating compliance effects on timely payments of the October 2020 property tax bill, along with forward and backward payments.⁹ We present visual evidence of the intervention’s effects in Figures 8 through 10, and we summarize the corresponding point estimates in Table 4.

Figure 8 shows the effect of the communication campaign on payment rates of pre- and post-intervention bills, for 24 consecutive monthly bills between January 2019 and December 2020. The top figures show payment rates for the control group (women in red and men in blue). The bottom figures report the treatment effects—i.e., the difference between treated and control units—and 95% confidence intervals. The left panels focus on timely payments, defined as bills paid before the 27th of the corresponding month (i.e., any payment made after the 27th is considered unpaid). Hence, pre-intervention bills mechanically exclude any past-due payment triggered by our intervention. The right panels consider on-time and late payments made until December 2020 and, thus, capture backward payments triggered by our intervention (e.g., individuals who, after receiving the letter, pay the October 2020 and previous unpaid bills).

Our analysis, as depicted in Figure 8 panel (a), indicates that men and women in the control group have similar tax payment rates. Approximately 50% of individuals in the control group pay the property tax bill on time during normal times, a figure that increases to about 60% when accounting for late payments. Interestingly, despite the significant drop in payment rates at the onset of the COVID-19 pandemic, there are no noticeable gender-based differences. This indicates that, in our setting, macroeconomic shocks seem to affect the tax behavior of men and women equally.

Additionally, Figure 8 panel (b) shows that men and women respond similarly to personalized property tax letters. The bottom left panel reveals that recipients of the letter are substantially more likely to pay the October 2020 treated bill on time compared to those in the control group, with no significant gender differences. Table 4 summarizes the intention-to-treat effects for both groups: payment rates of treated women are 4.2 percentage points higher than those for control

9. Appendix section C.3 confirms that our groups are balanced and comparable.

women, representing a 12% increase from the control baseline of 35%. For men, the increase is 4.7 percentage points, or a 13.5% rise over the control baseline. Reassuringly, we do not observe any differences in timely payments between treatment and control for pre-intervention bills.

Our study reveals that the impact of the personalized tax letters extended beyond the targeted October 2020 bill. The bottom right panel of Figure 8 provides compelling visual evidence that the letters encouraged some neighbors to clear past-due debt from previous billing periods, particularly following the implementation of COVID-19 lockdown measures in Argentina in April 2020. Despite receiving the letters in early October, the intervention also prompted some taxpayers to pay their November and December bills on time. Crucially, the magnitude of the responses was similar for both men and women.

We further exploit the detailed granularity of our property tax data to examine the payment dynamics of the October 2020 bill by gender, both before and after the due date. To this end, we first rectangularize the dataset using the payment date variable to obtain a daily balanced panel of payments. For each individual i , the payment indicator pay_bill_{id} is set to zero up until the day the payment is made, and equal to one thereafter. We then run the following fully saturated regression

$$pay_bill_{id} = \gamma_d + \sum_{d=Sept,25th}^{Oct,31st} \beta_d \cdot Treat_{id} + X'_{id} \cdot \rho + \epsilon_{id} \quad (2)$$

where γ_d are day indicators spanning September 25th to October 31st, $Treat_{id}$ is an indicator for those who received a letter, interacted with the day dummies, and X_{id} are controls. We run this regression separately for women and men and plot the coefficients together in Figure 9. Intuitively, β_d captures the cumulative difference in payment rates between the treated and untreated neighbors for each day d , providing insight into the dynamic effects of the intervention.

The top panel in Figure 9 presents the distribution of payment dates for the October 2020 bill among women and men, with most payments occurring before the due date on October 9th. The bottom panel reveals an immediate and statistically significant increase in the payment rate among treated units. Notably, men are more sensitive and respond earlier to the tax letter than women. The compliance effect for men emerges both numerically and statistically earlier than women, reaching a magnitude of about 5 percentage points. While men's treatment effect remains relatively constant after the due date (October 9th), treated women continue to make overdue payments, gradually narrowing the gender compliance gap.¹⁰

Lastly, we investigate heterogeneous responses across property value quintiles and by gender,

10. Figure C.8 in the appendix presents the analog of Figure 9 for the pre-treatment July 2020 bill. Reassuringly, the evidence indicates a zero pre-treatment effect on payment rates between treated and untreated groups for either men or women.

as illustrated in Figure 10. The top panel displays on-time payment rates for the control group, distinguishing between women (in red) and men (in blue). Tax compliance increases with property valuation but is otherwise similar for men and women. The bottom panel reports the treatment effects for each quintile—i.e., the difference between treated and control units—alongside 95% confidence intervals. While women present homogeneous responses across quintiles, men exhibit a slightly negative gradient. Notably, in the lowest quintile, the personalized tax letter boosted timely payments of men by 8 percentage points—twice the increase observed for women.¹¹

Taken together, our findings indicate that men and women exhibit similar payment rates and responses to personalized tax letters. However, some of our evidence suggests that men might be slightly more sensitive to this type of intervention. For instance, men tend to respond more quickly to tax letters, though women eventually catch up. Additionally, men with lower-valuation properties tend to react more strongly to the intervention. Our findings thus somehow downplay the potential importance of gender-dependent strategies to enhance property tax compliance.

6 Conclusion

The increased access to administrative data and tax authorities' willingness to engage in randomized experiments offer a new opportunity to shed light on the differential effects of taxation by gender. In the context of property taxation in the municipality of Tres de Febrero, Argentina, we first document large gaps in property ownership by gender, particularly among high-value properties, consistent with other settings where these gaps were investigated. Nonetheless, administrative records show that compliance, measured by timely payment of property tax liabilities, is very similar across properties with male and female owners. Furthermore, we exploit a randomized trial aimed at improving compliance and show that reactions to an enforcement letter are similar across genders.

The overall absence of gender differentials in baseline compliance and response to the intervention is surprising. Surveys have shown that in contrast to men, women tend to think that the tax code is fairer, the likelihood of getting caught for evasion is greater, and they overestimate the penalties for evasion (D'Attoma, Volintiru, and Steinmo, 2017; Torgler and Valev, 2010). In terms of behavior, a number of tax compliance lab experiments have also shown women to be more compliant than men (e.g., see Chung and Trivedi, 2003; Spicer and Hero, 1985). López-Luzuriaga and Scartascini (2023), studying a similar setting of property tax in another city in Argentina, document that women are more compliant at baseline (but similar to our result, responses to an

11. As a validation, Figure C.9 shows no effects for the September 2020 pre-intervention bill.

intervention are not different by gender).

Our evidence, however, suggests that, at least in the context of property taxes in Argentina, men and women seem to be equally compliant. One potential explanation is that the enforcement of property taxes in our context is weak. If the main reasons women are more compliant are beliefs about enforcement and penalties, and in our setting these are overwhelmingly low across genders, any compliance disparities might be muted. If, on the other hand, disparities are driven by higher tax morale among women, we would have expected larger gender differentials. On this last point, we note that the recent experimental study by Guerra and Harrington, [2018](#) finds that individual self-reported tax morale cannot predict actual evasion choices. Therefore, our results should encourage more investigation into whether gender differentials in tax compliance are widely observed in the field, as opposed to in lab experiments or self-reported surveys.

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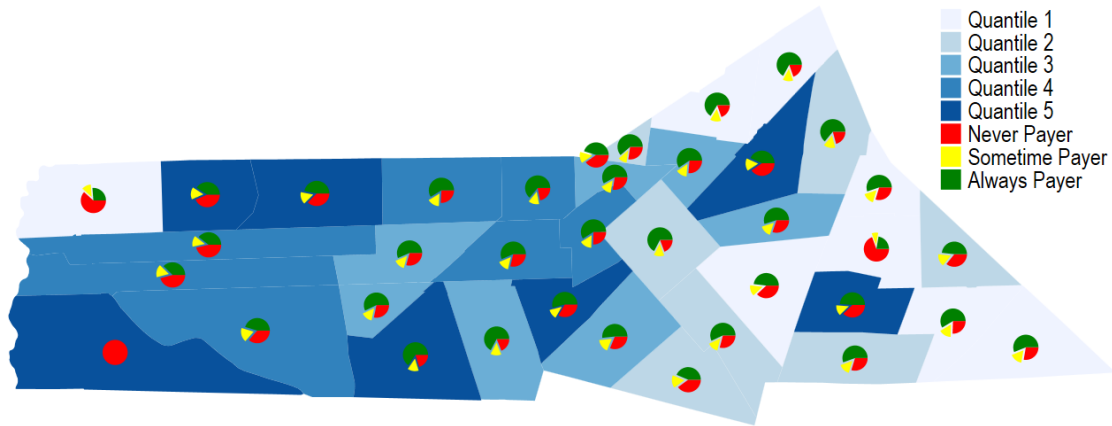
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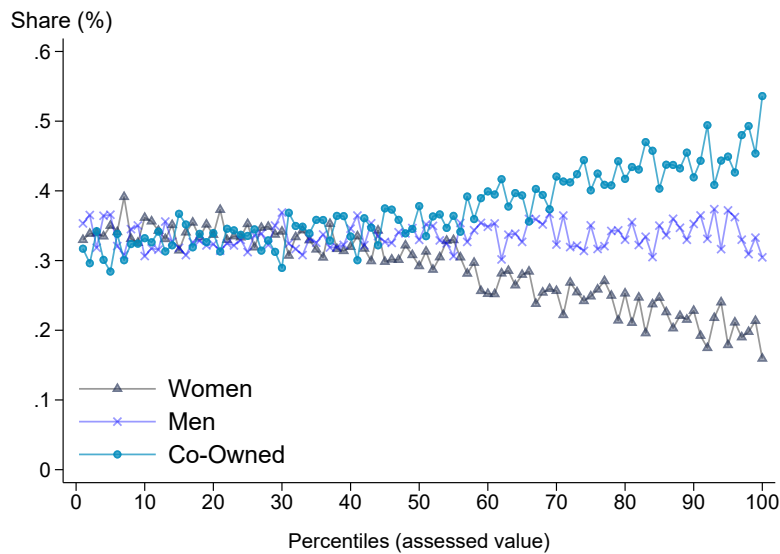
Figures

Figure 1: Property Assessed Value and Tax Compliance



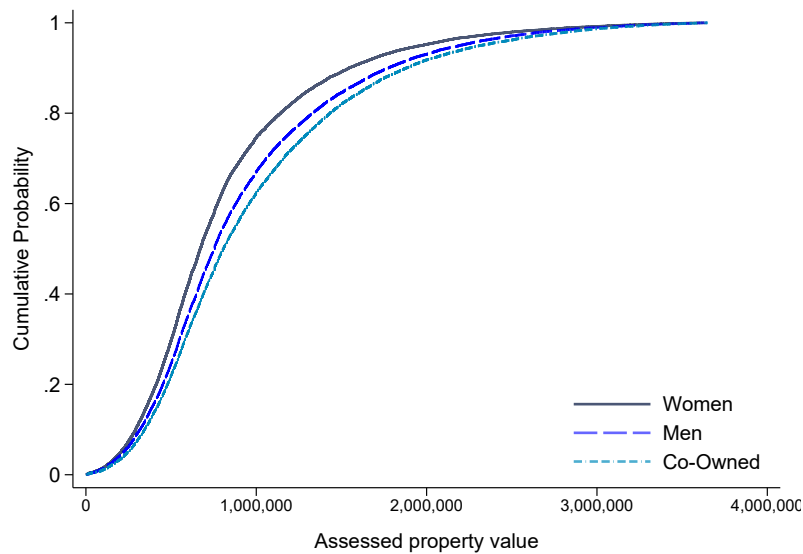
Note: This map illustrates the assessed property values (quantiles) for residential properties at the cadastral section level, determined by the 2018 ARBA valuation. Additionally, it visualizes compliance patterns of property taxpayers categorized as 'Always Payers,' 'Never Payers,' and 'Sometime Payers' based on administrative records from 2019 provided by the Tres de Febrero municipality

Figure 2: Residential ownership by gender and assessed value, 2018



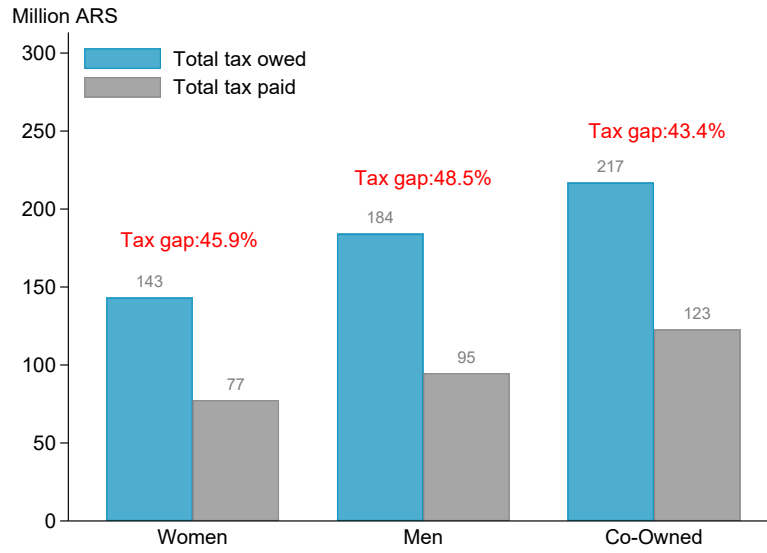
Note: This figure presents the share of residential properties across percentiles of assessed values for 2018 in *Tres de Febrero*. Sums are made at the percentile level. “Women” and “Men” are single-owned residential properties, while the “Co-owned” category includes residential properties reported under the ownership of men-women, men-men, or women-women.

Figure 3: Assessed property value by gender, 2018



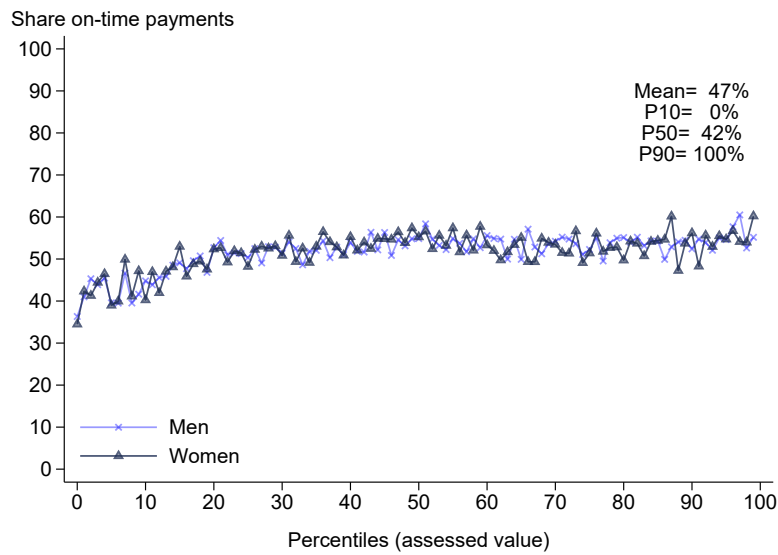
Note: This figure presents Cumulative Distribution Functions (CDF) for the assessed values of residential properties by gender, for 2018 in *Tres de Febrero* municipality. The X-axis corresponds to Argentinian Pesos and it was trimmed at the 99th percentile for visualization purposes.

Figure 4: Tax gap by gender, 2019



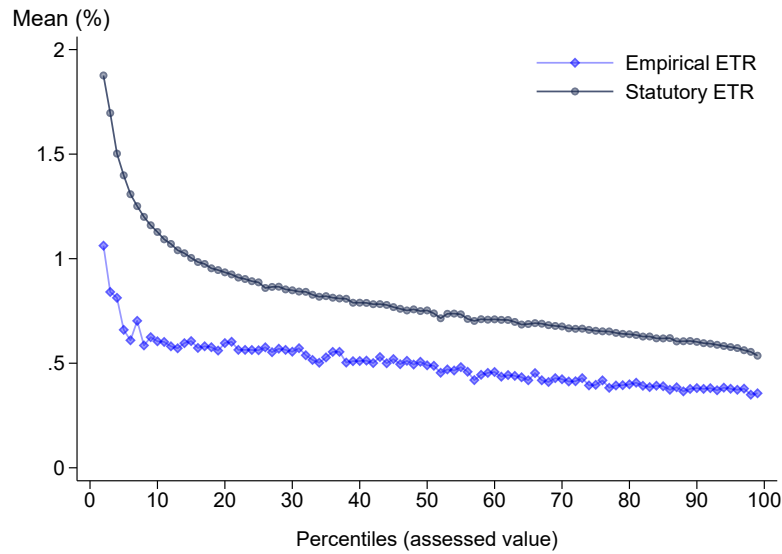
Note: This figure compares the aggregate property tax liabilities and tax effectively collected by gender in the *Tres de Febrero* district for the year 2019. Values are expressed in millions of Argentinian Pesos (ARS). The tax gap, calculated as $1 - (\sum_i \text{tax paid}) / (\sum_i \text{tax owed})$, is reported in red above the bars. These numbers are for residential properties only. The macro total tax owed in 2019 was ARS 544m, compared to a macro total tax paid of ARS 295m, yielding a tax gap of ARS 249m (46% of the tax owed). The tax gap is economically significant. It is roughly equivalent to the local expenditure on public safety of ARS 246m, representing about 8% of total spending (see Table A1).

Figure 5: Bills paid on-time, 2019



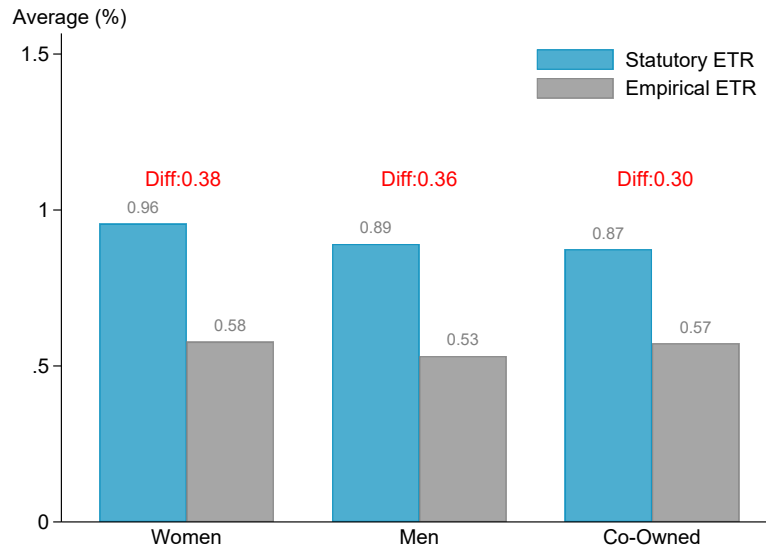
Note: This figure shows the fraction of bills paid on time in 2019, by gender, and across percentiles of assessed values in *Tres de Febrero* municipality for residential properties only. The conditional mean is built by collapsing the average of the individual fraction of payments (out of twelve in the year) paid on time. We define a bill paid on time as any bill registered up to the 27th day of the due month and bills paid in an annual payment at once.

Figure 6: Effective tax rates, 2019



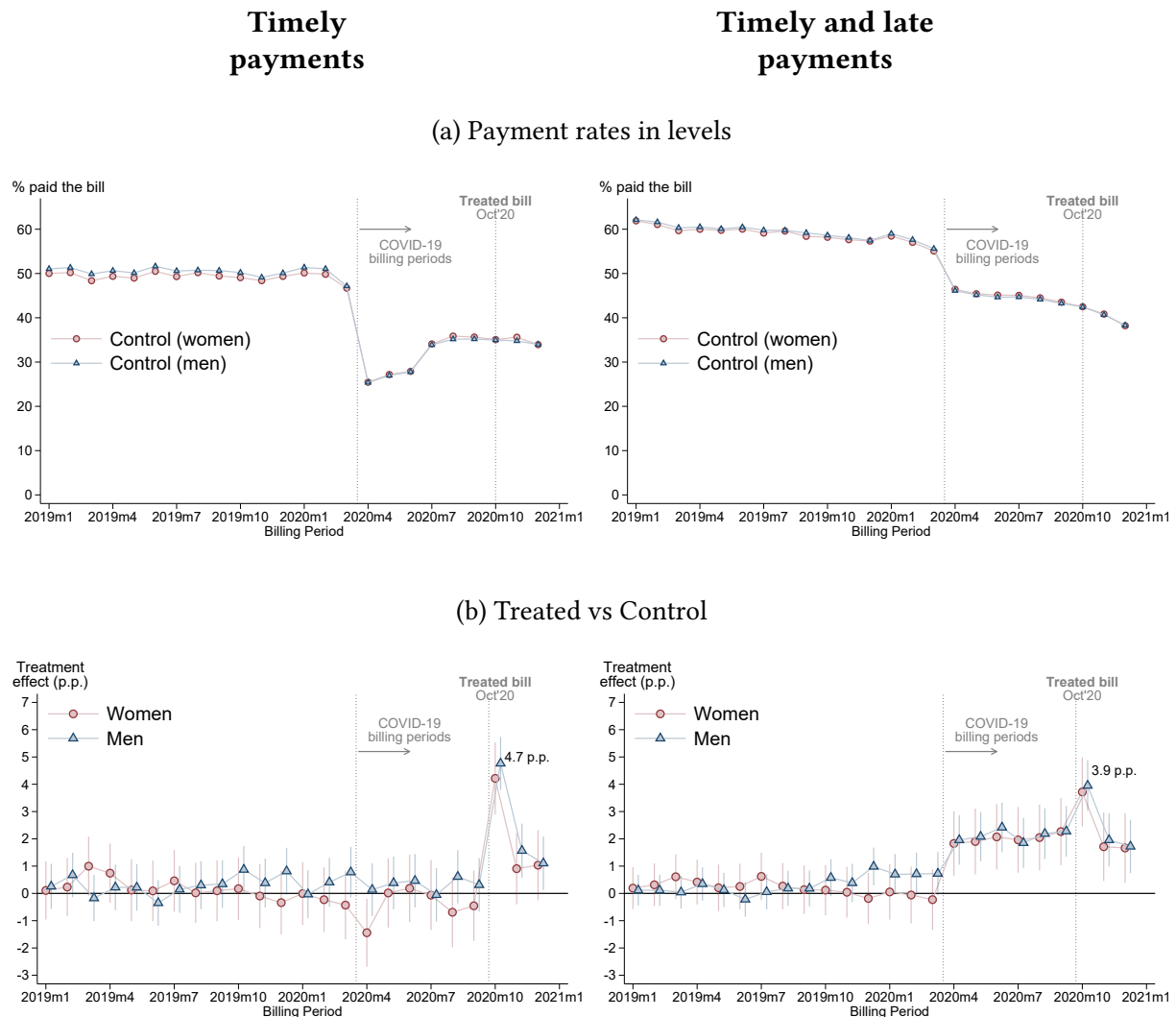
Note: This figure presents mean effective tax rates (ETR) for residential properties only, across percentiles of assessed property values in *Tres de Febrero* municipality for 2019. Dark navy dots report ETR as the tax liability to property value ratio, while blue dots report the ratio between paid taxes and property value. The figure does not include taxpayers who paid the annual tax liability in one go. We also exclude the first percentile for visualization purposes.

Figure 7: Effective tax rates, by gender, 2019



Note: This figure presents mean effective tax rates (ETR) in 2019 for residential properties only and across gender ownership. Dark navy bars report ETR as the tax liability to property value ratio, while blue dots report the ratio between paid taxes and property value. The figure does not include taxpayers who paid the annual tax liability in one go. We also exclude the first percentile for visualization purposes.

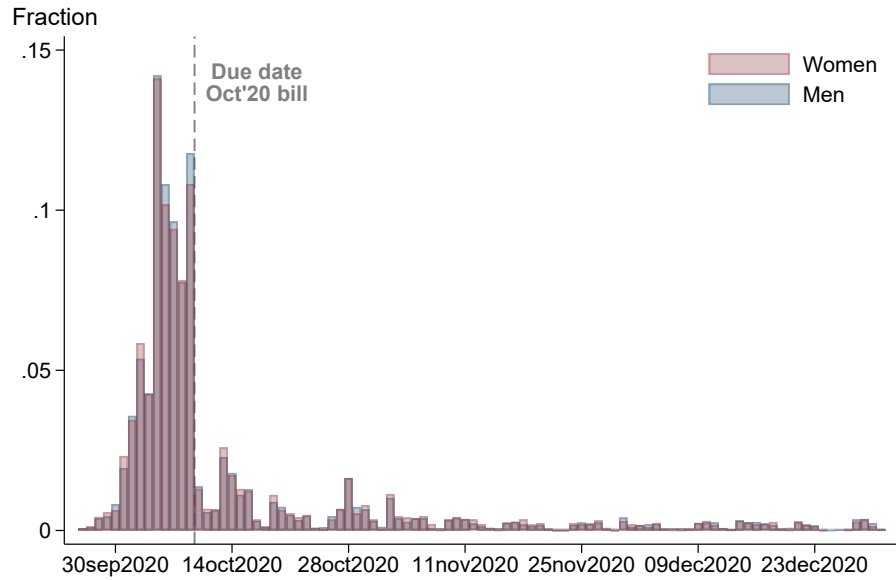
Figure 8: Property tax compliance: women and men respond similarly to personalized tax letters



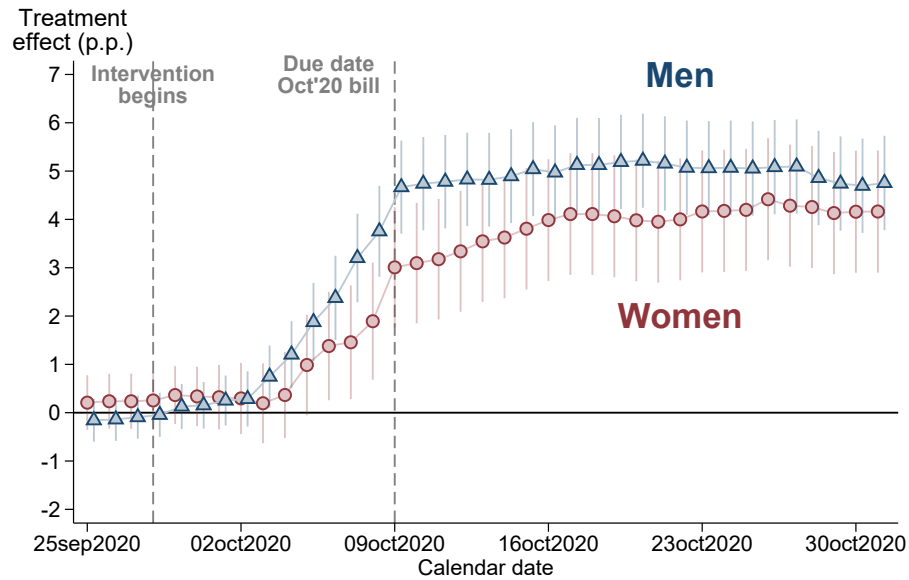
Notes: These figures show the effect of the communication campaign on payment rates of pre- and post-intervention bills, for 24 consecutive monthly bills between January 2019 and December 2020. The left panels only consider timely payments, defined as bills paid before the 27th of the corresponding month (i.e., any payment made after the 27th is considered unpaid). Hence, pre-intervention bills mechanically exclude any past-due payment triggered by our intervention. The right panels consider timely and past-due payments made until December 2020 and, thus, capture backward payments triggered by our intervention (e.g., individuals who, after receiving the letter, pay the October 2020 and previous unpaid bills). The top figures show payment rates in levels for the control group (women in red and men in blue). The bottom figures report the treatment effects—i.e., the difference between treated and control units—and 95% confidence intervals for the 24 billing periods. These regressions control for tax compliance in 2018, zip codes, an indicator for undelivered letters, and an indicator for those paying no bills in 2018. The letters were delivered between September 28th and October 7th, 2020. The vertical dashed lines denote the start of the COVID-19 pandemic in Argentina and the month the letters were delivered. Each coefficient is estimated in separate regressions. Standard errors are clustered at the street-block level. We find that men and women respond similarly to the treatment letter. The intervention also nudged some people to catch up with unpaid bills from April 2020 onward, but the increased tax compliance is the same for men and women.

Figure 9: Property tax compliance: men respond earlier to the letter, but women catch up

(a) Distribution of payment date for women and men (October 2020 bill)



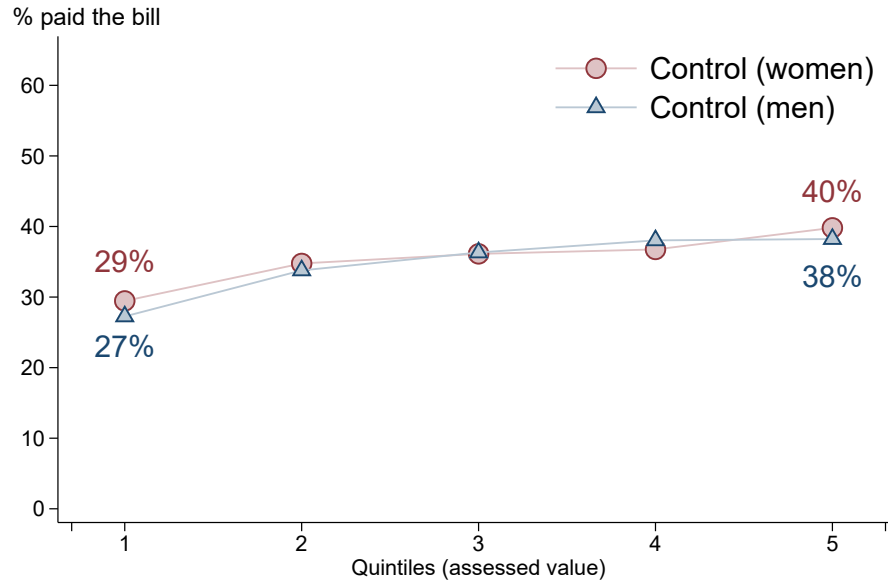
(b) Treated vs Control (October 2020 bill)



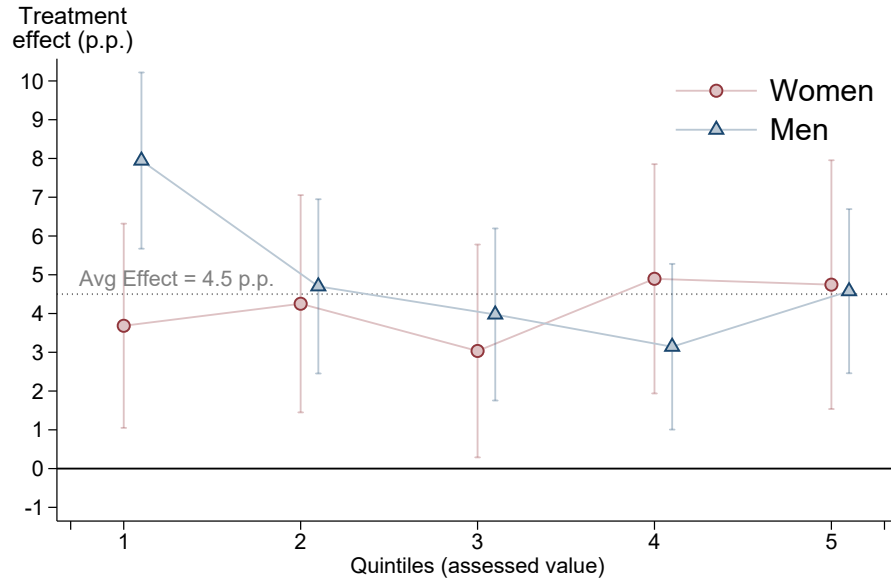
Notes: Panel (a) shows the fraction of men and women paying the October 2020 bill before and after the due date (October 9th, 2020). The area of each histogram integrates into one. A larger bar on a particular date means that the payment frequency of the corresponding group is higher than that of the other group. Panel (b) shows the coefficients and 95% confidence intervals from a saturated regression that computes, on each calendar day, the payment rate difference between treatment and control. Standard errors are clustered at the individual level. The first vertical bar denotes the start of the intervention. The second vertical bar indicates the due date for the October 2020 bill. The letters were delivered between September 28th and October 7th. While men respond earlier to the tax communication campaign, women eventually catch up.

Figure 10: Property tax compliance by quintiles and gender

(a) Timely payment rates by quintiles



(b) Treated vs Control



Notes: These figures show the effect of the communication campaign on payment rates by quintiles of assessed values. We focus on timely payments of the treated bill corresponding to October 2020. Timely payments are defined as bills paid before the 27th of the corresponding month (i.e., any payment made after the 27th is considered unpaid). The top panel shows payment rates in levels for the control group (women in red and men in blue). The bottom panel reports the treatment effects for each quintile—i.e., the difference between treated and control units—and 95% confidence intervals. The letters were delivered between September 28th and October 7th, 2020. Each coefficient is estimated in separate regressions. Standard errors are clustered at the street-block level. While women present homogeneous responses across quintiles, men exhibit a slightly negative gradient.

Tables

Table 1: Category of property and statutory tax rates (STR)

Category	N° Properties		STR by year (%)		
	N	%	2018	2019	2020
Residential	109,729	82.8	0.30	0.50	0.69
Residential with commerce/factory	10,073	7.6	0.45	0.75	1.04
Commercial	7,425	5.6	0.70	1.18	1.68
Empty lot	1,166	0.9	0.70	1.27	1.84
Industrial	3,699	2.8	0.70	1.22	1.77
Civil entities	220	0.2	0.10	0.17	0.24
Religious entities	170	0.1	0.10	0.17	0.24
Wholesale establishments	1	0.0	0.45	0.76	1.10

Note: This table reports statutory tax rates by category of property as defined in the tax code (*Ordenanza Impositiva*) every year. The first two columns display the number of properties in 2018. The tax base used to calculate the monthly fee corresponds to the assessed value in 2018 provided by the provincial tax authority (ARBA). Despite an annual inflation of 47% in 2018, 54% in 2019, and 36% in 2020, this base remained unadjusted. To compensate, the municipality increased the tax rates.

Table 2: Summary statistics, all properties

	2018	2019	2020
Panel A: Property Characteristics			
Residential property	0.83	0.83	0.83
Property located in Caseros	0.30	0.30	0.30
Linear front meters	974.52	985.09	986.00
Residential properties with gender assigned	0.74	0.74	0.74
Female-owned residential properties	0.21	0.21	0.21
Co-owned residential properties	0.28	0.28	0.28
Panel B: Taxation and Payments			
Assessed property value (AR\$)	1,180,816.32 (5,258,373.90)	1,177,533.91 (5,297,205.89)	1,180,473.90 (5,356,855.64)
Tax liability (AR\$)	7,702.69 (49,221.29)	10,101.05 (60,695.06)	13,744.62 (92,778.56)
Statutory ETR (%)	0.73 (1.04)	0.96 (1.20)	1.25 (1.44)
Empirical ETR (%)	0.46 (0.62)	0.59 (0.79)	0.63 (0.94)
Payment to tax liability ratio (%)	63.91 (45.17)	62.15 (45.91)	50.69 (46.13)
Paid on time	0.68 (0.47)	0.66 (0.47)	0.63 (0.48)
Always payer	0.55 (0.50)	0.54 (0.50)	0.40 (0.49)
Never payer	0.28 (0.45)	0.31 (0.46)	0.36 (0.48)
Number of properties	132,483	133,422	133,690

Note: This table reports summary statistics at the yearly level. While the core analysis in this document focuses on residential properties only, in this table, we include the universe of properties in *Tres de Febrero* municipality, that is, residential, commercial, and industrial properties, empty lots, civil or religious entities, and wholesale establishments. **Panel A** displays averages. **Panel B** displays means and standard deviations (in parenthesis). The last panel includes counts of unique properties (*número de cuenta*).

Table 3: Payment methods of residential bills by gender, 2018

	$\mathbb{P}(\text{Cash}=1)$		$\mathbb{P}(\text{Online}=1)$		$\mathbb{P}(\text{Mixed}=1)$		Share Online	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
1(Female=1)	0.019*** (0.00)	0.017*** (0.00)	-0.017*** (0.00)	-0.013*** (0.00)	-0.002 (0.00)	-0.003 (0.00)	-0.020*** (0.00)	-0.016*** (0.00)
Constant	0.583*** (0.00)	0.688*** (0.01)	0.162*** (0.00)	0.076*** (0.00)	0.241*** (0.00)	0.214*** (0.00)	0.269*** (0.00)	0.146*** (0.00)
N	62,390	57,379	62,390	57,379	62,390	57,379	62,390	57,379
R-Squared	0.00	0.15	0.00	0.12	0.00	0.12	0.00	0.14
Mean Dep Var 2018	0.59	0.58	0.16	0.16	0.24	0.24	0.26	0.27
Control for age?	No	Yes	No	Yes	No	Yes	No	Yes
Block FE?	No	Yes	No	Yes	No	Yes	No	Yes
Value decile FE?	No	Yes	No	Yes	No	Yes	No	Yes

Note: This table reports differences in means from linear probability models for the likelihood of paying bills using cash (if the taxpayer paid 100% of its bills using cash through any bank or in person at any municipality account), online methods (if the taxpayer paid 100% of its bills using *Mercado Pago*, *Pago Link*, or *Pagos Mis Cuentas*), or a mix, respectively. “Share online” is the total tax paid using online methods. The sample is restricted to residential properties and payments made in 2018. “Never payers” are excluded from estimations. Columns titled with even numbers include fixed effects for property blocks and deciles of assessed property values and also include as controls the age and the age squared of the property owner. The gender corresponds to the property owner, not the gender of the person making the payment. Robust standard errors in parenthesis.

Table 4: Treatment effects on property tax timely payments for women and men

Dependent variable: Pr(pay the bill)	Placebo bill (Sept'20)		Intervention bill (Oct'20)	
	Women	Men	Women	Men
	(1)	(2)	(3)	(4)
Treated	-0.45 (0.66)	0.31 (0.50)	4.22*** (0.67)	4.76*** (0.49)
Control mean	35.64	35.20	35.08	34.99
Observations	18,089	30,399	18,089	30,399
Number of clusters (blocks)	3,813	3,957	3,813	3,957

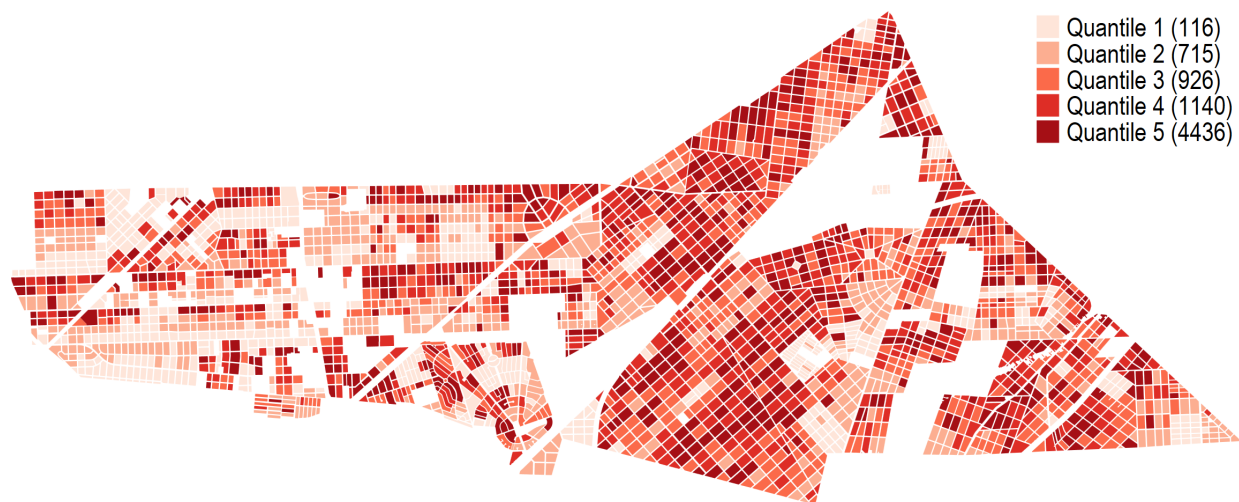
Notes: This table summarizes the treatment effects of the randomized tax communication campaign on property tax payments for men and women. Each column corresponds to a separate regression. The dependent variable is an indicator for paying the bill on time. Timely payments are defined as bills paid before the 27th of the corresponding month (i.e., any payment made after the 27th is coded as unpaid). Columns 1 and 2 correspond to the pre-intervention bill of September 2020; Columns 3 and 4 correspond to the October 2020 intervention bill. The estimates match those shown in Figure (8). The letters were delivered between September 28th and October 7th. The due date for the October 2020 bill was October 9th. The row *Control mean* displays the average payment rate for untreated units who received no letter. These regressions control for tax compliance in 2018, zip codes fixed effects, an indicator for undelivered letters, and an indicator for those paying no bills in 2018. Standard errors clustered by blocks are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

SUPPLEMENTARY MATERIALS FOR: “EXPLORING THE GENDER DIVIDE IN REAL ESTATE OWNERSHIP AND PROPERTY TAX COMPLIANCE”

A Residential ownership in Tres de Febrero

Based on the 2021 ARBA cadaster, the municipality boasts approximately 140,000 registered properties. As depicted in [Figure A.1](#), property density peaks in the southern areas, with an average of over 500 to 1,000 properties per block in many regions. In contrast, the northern areas exhibit lower density, with around 100 to 200 properties per block. Commercial, service, and manufacturing properties are predominantly clustered in the central areas and corridors of the municipality, particularly in Caseros, where close to 30% of all properties are concentrated, and Ciudadela towns, where they coexist with residential zones. Of the total properties registered, approximately 83% are categorized as residential, while 8% are categorized as residential with commercial or industrial purposes, 6% are purely commercial, and the remaining 3% fall into various other categories.¹

Figure A.1: Number of residential properties

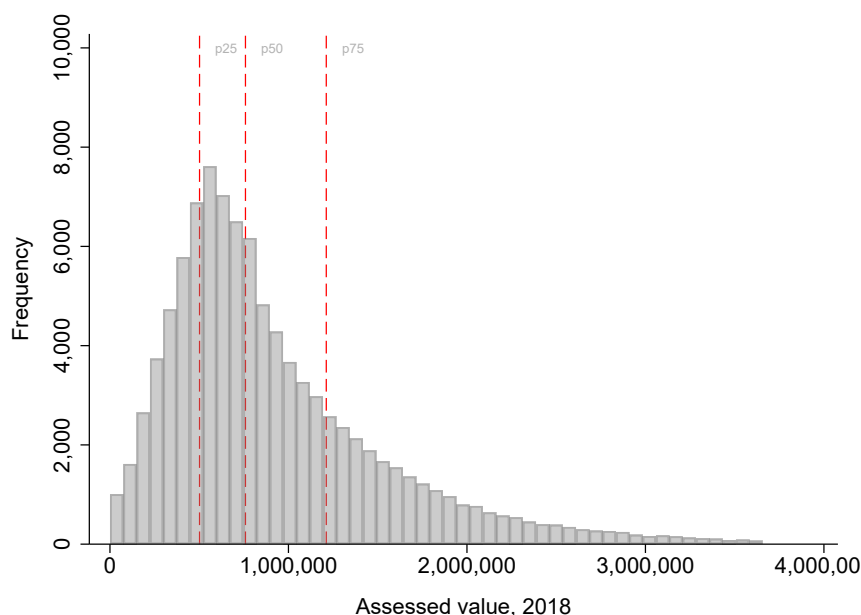


Note: This map presents the number of residential properties at the block "manzana" Level in Tres de Febrero 2021 ARBA register.

1. These other categories encompass empty lots, industrial, civil and religious entities, as well as wholesale establishments.

Focusing on residential properties only, in 2018, assessed values spanned from 1,800 AR\$ to $\approx$ 4 million AR\$ in 2018 based on the municipality estimates. The median assessed value falls at $\approx$ 759,000 AR\$, while the mean residential property assessed value is 979,000 AR\$, as shown in [Figure A.2](#). When examining residential property ownership patterns across genders, as illustrated in [Figure 2](#), similarities emerge in the ownership distribution up to the 25th percentile of property values, with women, men, and jointly-owned properties encompassing a range of 30%-40%. However, as property values ascend beyond the 50th percentile, a noticeable trend unfolds: women's total ownership percentage begins to diminish, dwindling to around 20% of total properties at the highest decile, and further declining to 15% at the 99th percentile. In contrast, men maintain a relatively stable ownership share, hovering between 30% and 35% of total properties at the upper levels of property value distribution. Concurrently, jointly-owned properties experience a surge from the 50th percentile onwards, representing approximately 55% of all properties at the top of assessed values.

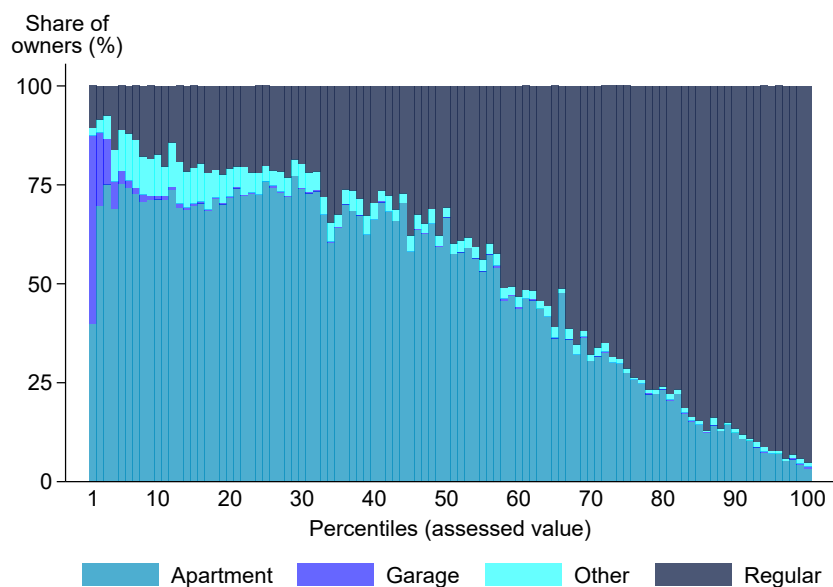
Figure A.2: Assessed values for residential properties, 2018



Note: This figure presents the histogram with the assessed value of residential properties for 2018 in *Tres de Febrero*. These values are provided by the provincial tax authority (ARBA). The vertical red lines show the 25th, 50th, and 75th percentiles, respectively. In the Y-axis we report the number of properties, while X-axis corresponds to Argentinian Pesos (AR\$). The distribution was trimmed at the 99th percentile for visualization purposes.

In our examination of the distribution of residential ownership categories across property values, depicted in Figure A.3, we can see that at lower assessed values, there is a predominant presence of garages and apartments, with the latter comprising a substantial portion ranging from 40% to 74%, while the remainder falls under the categories of regular properties or others. However, this distribution inverts as property values escalate. Near the top decile of property assessed values, merely around 15% represent apartments, with the majority being categorized as regular properties.

Figure A.3: Distribution of residential ownership, 2018



Note: This figure presents the number of residential properties across percentiles of assessed values for 2018 in *Tres de Febrero*. Category “other” includes mono-blocks, which are social housing. The category “regular” refers to single units such as houses or shops.

B Coding sex assigned at birth

Determining the assigned sex at birth in administrative tax data is usually a difficult task. Gender is typically not reported on most tax forms. A solution used by scant literature is to infer gender using names and text analysis (e.g., see Scot et al., 2023; López-Luzuriaga and Scartascini, 2023, for recent applications). In this paper, we rely instead on personal tax identifiers known as CUIT (an acronym for *Clave Única de Identificación Tributaria*, “unique tax identification”), which allows us to infer the gender assigned at birth from the first two digits and the last digit. The CUIT is a national tax identification number in Argentina, assigned by the Federal Administration of Public Revenue (AFIP) to people and businesses.

The CUIT number has a **YY-XXXXXXXX-Z** structure, with the following meaning:

- **YY** is the type: 20, 23, 24, 27 (individuals), 30, 33, 34 (businesses).
- **XXXXXXXX** is the national ID number (DNI) or business number.
- **Z** is a verifying digit.

In the case of individuals, the following administrative rules can be used to determine sex:

- **Female:** YY=27, or YY=23 and Z=4.
- **Male:** YY=20, or YY=23 and Z=9.
- **Unknown:** YY=24. This applies to a few cases for which we will have missing gender.

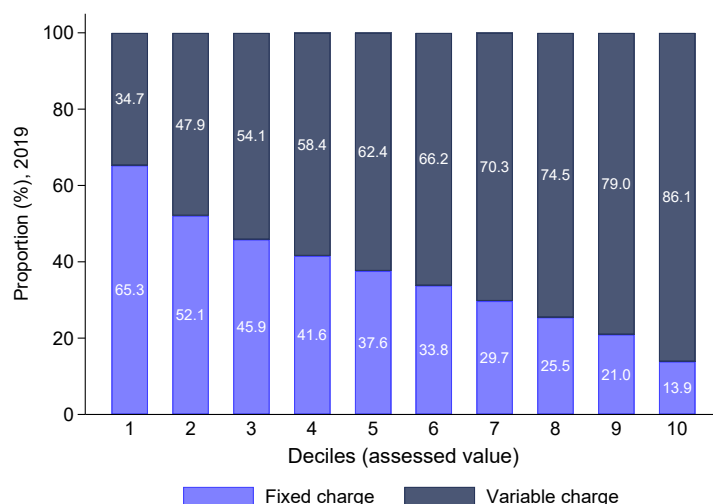
Following methodological guidance from our team, the municipality’s staff utilized this structural feature of the CUIT in the property register, to infer the gender assigned at birth and the owners’ ages for over 100,000 properties annually. We then linked the gender indicator to the main property tax database to generate a sex-disaggregated analysis.

Following the strategy described above, we were able to correctly assign the gender to 74% of the 108,321 residential property owners using data provided by ARBA from 2018, while the remaining 26% were set to missing.

C Additional material

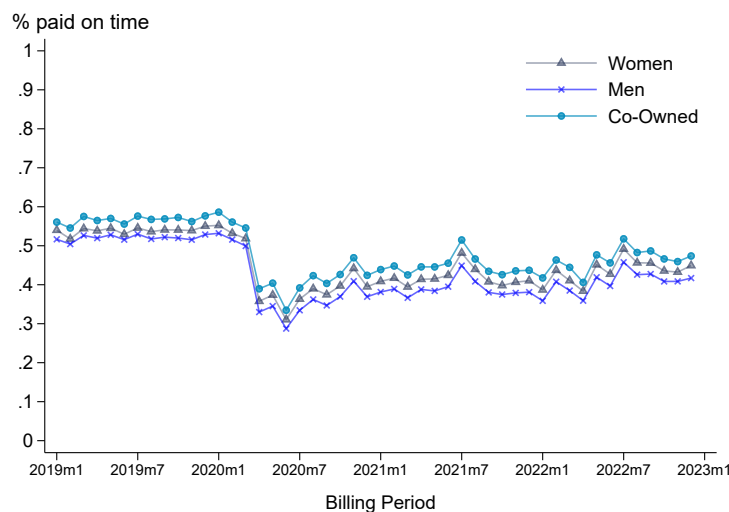
C.1 Summary Statistics

Figure C.4: Fixed and Variable Charges of the Property Tax Bill, by Deciles



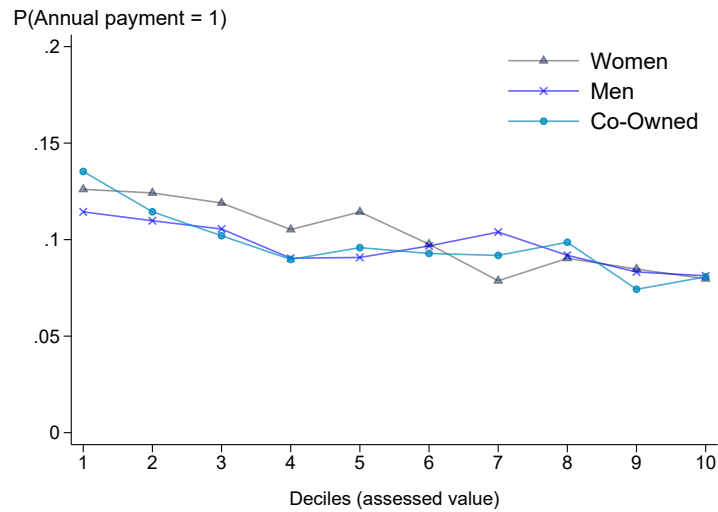
Note: This figure shows the relative importance of variable and fixed components of the property tax bill by deciles of assessed values in 2019. According to legislation in 2019, variable charges for residential properties were 0.5% of assessed value, and fixed charges amounted to a flat fee of AR\$ 167 (AR\$ 90 for security services, AR\$ 52 for maintenance services, and AR\$ 25 for health services).

Figure C.5: Dynamics for payments on-time



Note: This figure presents a time series for the fraction of residential bills that were paid on time. We define a bill paid on time as any bill that was registered up to the 27th day of the due month and bills that were paid in an annual payment at once.

Figure C.6: Annual payments by gender, 2019



Note: This figure presents conditional means for annual bills made in one payment, across percentiles of assessed property values in 2019. Only residential properties are included.

Table A1: Breakdown of Local Government Expenditures in 2019

Type	Total Spending (mill pesos)	% of Total
1. Waste collection/management	933M	30.1
2. Healthcare	286M	9.2
3. Public safety	246M	7.9
4. Education and culture	154M	5.0
5. Maintenance and cleaning work	124M	4.0
6. Social programs	121M	3.9
7. Infrastructure	106M	3.4
8. Parks and public spaces	85M	2.7
9. Street/outdoor lighting + traffic signals	74M	2.4
10. Other	973M	31.4
Total	3104M	100

Note: This table reports local government expenditures in Tres de Febrero in 2019. It encompasses the total spending by local authorities in providing public services and making capital investments. Expenditure in 'Other' category includes general administration (e.g., salaries), IT services, communication services), debt services, sports and other recreational activities, etc.

C.2 Additional Material from the RCT

Figure C.7: Example of the intervention letter



Tus impuestos municipales ahora vienen en la BOLETA DIGITAL

ID: XXXXX

TITULAR: _____

DIRECCIÓN: CAP. MADARIAGA N° _____ LOCALIDAD: 11 de Septiembre

C.P.: 1657

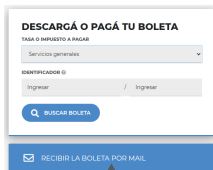
PARTIDA: XXXXX/7

Te queremos contar que ahora en Tres de Febrero tu boleto municipal de la Tasa por Servicios Generales (TSG) es 100% digital. O sea, ya no se usa más el papel. Podés acceder a ella y pagarla desde el celular o la computadora. De esta manera, nos cuidamos entre todos al reducir la circulación y también cuidamos el medio ambiente. Es una situación difícil y te agradecemos el esfuerzo que estás haciendo para estar al día con tus impuestos, porque eso se transforma directamente en obras y servicios que no paran en tu barrio. Te informamos el estado de tu cuenta y te mostramos lo fácil que es:

PARTIDA: XXXXX/7	
Cuota 10 vencimiento 10 de octubre 2020:	347,29
Deuda año en curso*:	1.702,58
Deuda años anteriores*:	289,54
* Al 15/09/2020	

¿CÓMO PAGAR?

Ingresando a tasas.tresdefebrero.gov.ar completá los datos:



CLICKEÁ ESTE BOTÓN
y recibís todos los meses en tu mail.
También podés entrar a miboleta.tresdefebrero.gov.ar

1) Podés pagar ONLINE con



mercado pago

→ En el momento desde nuestra web.



¡PagoMis Cuentas

→ Obteniendo el código de pago electrónico para pagar desde la plataforma de tu banco o cajero automático.

2) Podés pagar en EFECTIVO en



rapi pago

→ DESCARGALA o llevá tu NÚMERO DE PARTIDA.

Por dudas comunicate con nosotros a reclamos.mistasas@tresdefebrero.gov.ar
Si esta carta llegó por error a tu domicilio, informanos en ese mismo correo electrónico

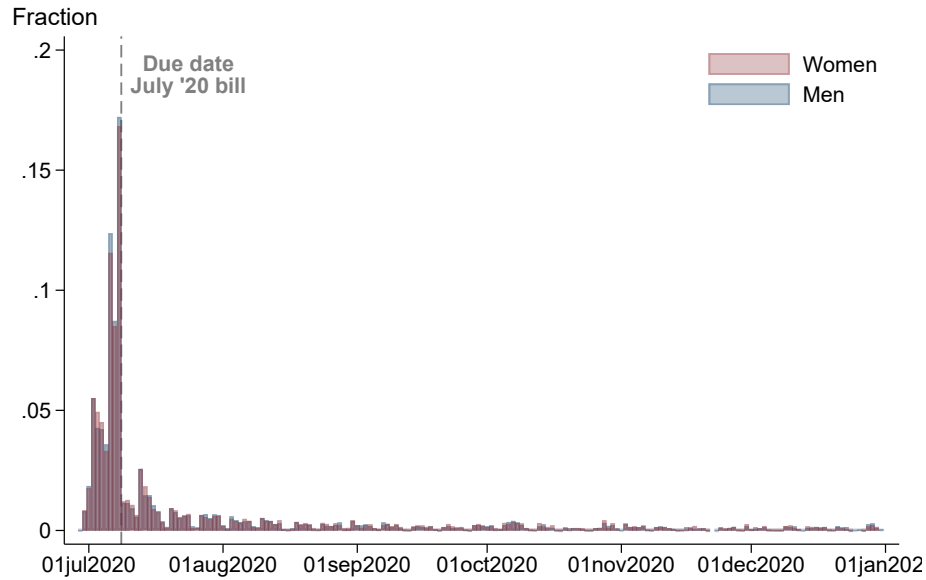
¡Muchas gracias!

Notes: This figure shows an anonymized example of the letters sent during the intervention between September 28th and October 7th, 2020. The headline reads: “Your municipal taxes are now available on the electronic bill.” The information below the headline contains the account holder’s name, address, and account number. The main text of the letter reads: “We would like to tell you that now in Tres de Febrero your municipal General Service Fee (TSG) bill is 100% digital. In other words, paper is no longer used. You can access it and pay for it from your cell phone or computer. In this way, we take care of each other by reducing circulation and we also take care of the environment. It is a difficult situation and we appreciate the effort you are making to keep up with your taxes, because that translates directly into constructions and services that do not stop in your neighborhood. We inform you of the status of your account and show you how easy it is:” The table below this text shows the account number, the amount due in the October 2020 billing period, the amount of past due debt from previous months of 2020, and the amount of past due date from earlier years. The large box below the table explains: (1) how to sign up for electronic billing, and (2) how to pay the bill and the different means of payment (online or in person). Finally, below the box, the text reads: “For questions, contact us at reclamos.mistasas@tresdefebrero.gov.ar. If this letter arrived mistakenly at your address, inform us in that email. Many thanks!”

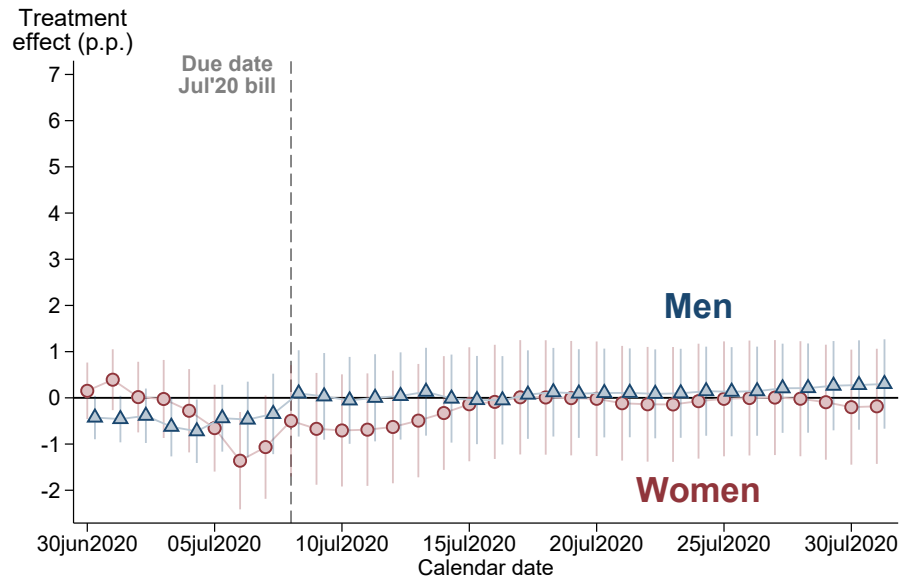
C-7

Figure C.8: Property tax compliance: no effect for the placebo bill of July 2020

(a) Distribution of payment date for women and men (July 2020 bill)



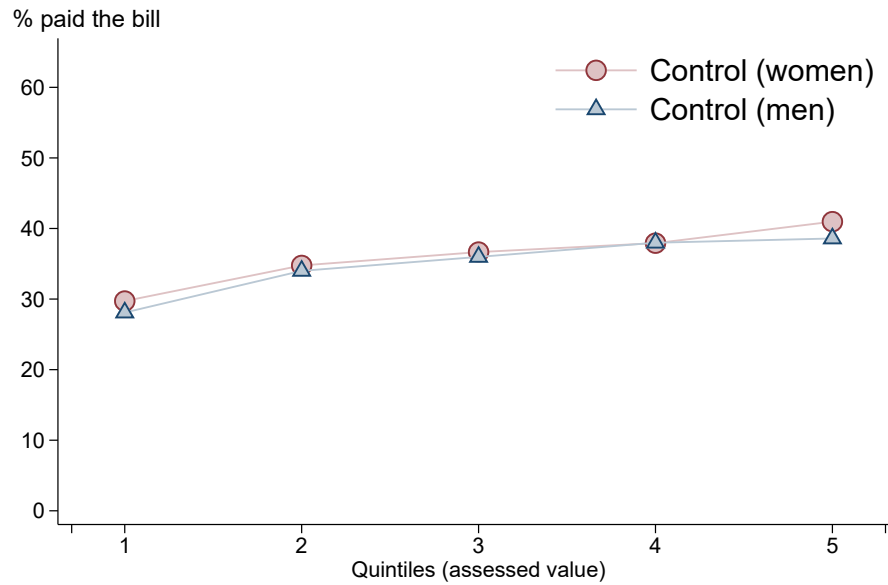
(b) Treated vs Control (July 2020 bill)



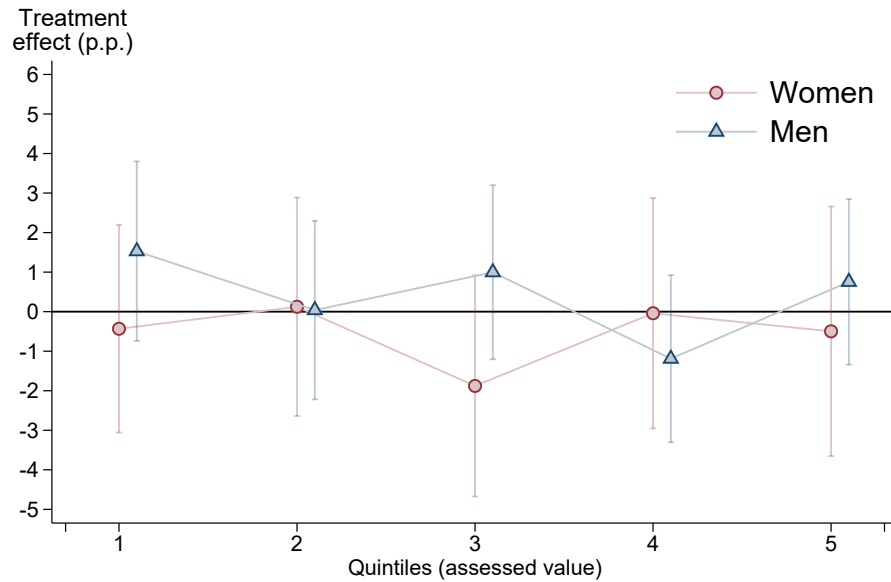
Notes: Panel (a) shows the fraction of men and women paying the July 2020 bill before and after the due date (July 8th, 2020). The area of each histogram integrates into one. A larger bar on a particular date means that the payment frequency of the corresponding group is higher than that of the other group. Panel (b) shows the coefficients and 95% confidence intervals from a saturated regression that computes, on each calendar day, the payment rate difference between treatment and control. Standard errors are clustered at the individual level. The vertical bar indicates the due date for the July 2020 bill. Reassuringly, and unlike Figure 9, men and women exhibit no response for a pre-intervention bill.

Figure C.9: Property tax compliance by quintiles and gender (PLACEBO)

(a) Timely payment rates by quintiles



(b) Treated vs Control (September 2020)



Notes: These figures show the placebo effect of the communication campaign on payment rates by quintiles of assessed values. We focus on timely payments of the September 2020 pre-intervention bill. Timely payments are defined as bills paid before the 27th of the corresponding month (i.e., any payment made after the 27th is considered unpaid). The top panel shows payment rates in levels for the control group (women in red and men in blue). The bottom panel reports the treatment effects for each quintile—i.e., the difference between treated and control units—and 95% confidence intervals. The letters were delivered between September 28th and October 7th, 2020. Each coefficient is estimated in separate regressions. Standard errors are clustered at the street-block level. This placebo exercise using a pre-intervention bill, confirms a null effect for men and women across quintiles of assessed values.

C.3 Balance checks

We run balance test checks to verify the comparability of the treated and control groups in terms of demographic and account-related characteristics in 2019, the year before the intervention. We estimate the difference in means with the following regression:

$$X_{ig} = \alpha + \theta \mathbf{1}(Treated_{ig}) + \varepsilon_{ig} \quad (3)$$

where X_{ig} is a property/account i characteristic observed in the data. We allow ε_{ig} to be correlated within clusters g (street-blocks) and use a cluster-robust variance estimator. In this regression, θ captures the average difference of X_{ig} of treated units relative to control units not receiving any letter. The results are reported in Table A2 and reassuringly confirm that our groups are highly balanced. The null effect on timely payments (i.e., excluding past-due payments) of the September 2020 bill—the bill prior to our intervention— sheds further light on the balance between groups (see Figure C.9).

Table A2: Balance tests

	Property Value	Front Meters	House type	Past due debt	Bill amount	N Bills paid 2019	Digital payment
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
A. Women and Men							
Treated	−0.00 (0.01)	−7.89 (10.13)	−0.00 (0.00)	−0.01 (0.01)	2.86 (5.31)	0.08 (0.06)	−0.01 (0.01)
Mean Control	13.65*** (0.01)	822.26*** (6.88)	0.92*** (0.00)	0.57*** (0.00)	386.25*** (3.66)	7.06*** (0.04)	0.38*** (0.00)
Observations	47,645	48,488	48,488	48,488	48,488	48,488	28,194
B. Women							
Treated	0.00 (0.01)	−11.87 (13.14)	−0.00 (0.00)	0.00 (0.01)	−0.20 (7.32)	0.02 (0.09)	−0.00 (0.01)
Mean Control	13.58*** (0.01)	779.42*** (9.30)	0.93*** (0.00)	0.56*** (0.01)	362.04*** (4.70)	7.02*** (0.06)	0.37*** (0.01)
Observations	17,766	18,089	18,089	18,089	18,089	18,089	10,416
C. Men							
Treated	−0.00 (0.01)	−5.86 (11.27)	−0.00 (0.00)	−0.01 (0.01)	4.49 (6.60)	0.11 (0.07)	−0.01 (0.01)
Mean Control	13.70*** (0.01)	847.87*** (7.23)	0.91*** (0.00)	0.57*** (0.00)	400.72*** (4.40)	7.08*** (0.05)	0.39*** (0.01)
Observations	29,879	30,399	30,399	30,399	30,399	30,399	17,778

Notes: This table shows balance regressions to test for differences in observable characteristics between the treatment and control groups. Each column corresponds to a separate regression (equation (3) in the text). The dependent variables in each column are: (1) the log of assessed property value; (2) the front meters of the property; (3) an indicator for the property being a house versus a house with a store; (4) an indicator for past due debt; (5) the amount paid in the bill corresponding to December 2019 (including zeroes); (6) the number of bills paid in 2019 (the maximum is 12); (7) for those who paid, whether they did so digitally. The rows *Mean Control* display the constant of each regression, corresponding to the average of the dependent variable for accounts that did not receive a letter. Standard errors clustered by blocks are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$